

INTERNAL REFERENCE · GxP COMPLIANCE

# GxP Compliance Reference

SiMPL Electronic Batch Records on SAP PI-PCS

Regulatory framework, SAP best-practice model, and how our XStep building blocks satisfy the controls

**Prepared:** 22 July 2026 · **Owner:** Carlos Redekopp (ISS)

**Scope:** AZ Phase 2 / Phase 3 EBR engagements — SiMPL eBR, SAP PI-PCS XSteps, functional/technical specs.

Orientation / mapping document — not a validation deliverable and not an executed record. Regulatory content is summarized from public standards and SAP sources (see Sources); confirm against the controlling regulation text and your site's Validation Master Plan before use in a formal submission.

## 1 Purpose & scope

This document orients our EBR work in the GxP compliance landscape: what the regulations require, how SAP's process-manufacturing model and best practices are structured, and — most importantly — **how the XStep building blocks we author already deliver the required controls**.

It is written for the SiMPL eBR engagements (AZ Phase 2 / Phase 3): scanning a paper Manufacturing Batch Record (MABR) into XStep mock-ups, writing functional/technical specifications, and building the real SAP PI-PCS XSteps that render the operator-facing PI Sheet. Use it as a shared reference when explaining *why* an XStep is designed the way it is, and as the seed for site-specific compliance documentation.

## 2 Our world in one page

We transform a static, sign-on-paper batch record into a validated, digital **Electronic Batch Record (EBR)** assembled from reusable **XStep** building blocks that run on SAP's process-manufacturing engine (**PI-PCS**).

| Stage              | Artifact                                  | What it is                                                                                                             |
|--------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Author-time (us)   | Mock-ups (PNG) + assembled EBR (PDF/HTML) | One card per XStep showing how it renders; the "New vs Old" PI Sheet cross-referenced to the paper MABR.               |
| Author-time (us)   | XStep Design Specifications (.docx)       | 13-section spec incl. Function Module(s) + Pseudocode; the buildable contract.                                         |
| Author-time (us)   | PI Sheet Sandbox records (JSON)           | Concrete JSON form of each mock-up published to the sandbox API.                                                       |
| Build (SapFractal) | SAP XStep (cmxsvn)                        | Real PI-PCS XStep cloned from the AZ Template; params, typed table columns, FMs, process messages, signatures.         |
| Runtime (SAP)      | PI Sheet                                  | Operator fills it in; SAP collects e-signatures, fires process messages (goods issue), writes back to MES / inventory. |

**The PI-PCS runtime chain:** Master Recipe (MBR) → Process Order → Control Recipe → PI Sheet → Process Instructions / XSteps → Process Messages. Recipe-specific values (targets, tolerances, headers) are overlaid onto the XStep's `IV_*` input parameters by the MBR at runtime.

## 3 The regulatory framework

Four instruments govern electronic records in a GxP manufacturing system. Our EBR must satisfy all of them.

| Instrument                | Scope                                                          | Core requirements that touch the EBR                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FDA 21 CFR Part 11</b> | Electronic records & electronic signatures (US, FDA-regulated) | Secure, computer-generated, <b>time-stamped audit trails</b> ; unique user IDs; secure, attributable <b>e-signatures</b> on critical actions; record accuracy, retrieval, and copies. |

|                         |                                                    |                                                                                                                                                                                                                             |
|-------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EU GMP Annex 11</b>  | Computerised systems (EU GMP)                      | Risk management; <b>validation</b> & fitness for intended use; data integrity; audit trails; access control; change & configuration management; business continuity.                                                        |
| <b>EU GMP Chapter 4</b> | Documentation                                      | Controlled, versioned master documents; contemporaneous recording; retention.                                                                                                                                               |
| <b>GAMP 5 (2nd ed.)</b> | Risk-based CSV guidance (ISPE — guidance, not law) | Categorises SAP as a <b>configurable product (Category 3–4)</b> → validation effort concentrates on <b>configuration &amp; customisation</b> , not standard delivered functionality; critical-thinking, risk-based testing. |

## ALCOA+ — the data-integrity yardstick

Every record the EBR captures should be: **A**ttributable, **L**egible, **C**ontemporaneous, **O**riginal, **A**ccurate — plus **C**omplete, **C**onsistent, **E**nduring, **A**vailable. §7 shows how specific XStep design choices map to each.

## 4 Computer System Validation (CSV) lifecycle

GMP-regulated functionality must be validated **before production use**. The V-model:

| Phase         | Deliverable                                  | Our contribution                                                                                     |
|---------------|----------------------------------------------|------------------------------------------------------------------------------------------------------|
| Requirements  | <b>URS</b> — User Requirements Specification | The paper MABR + client requirements workbook.                                                       |
| Design        | Functional / Design Specification            | Our <b>XStep Design Specifications</b> (functional + FM/Pseudocode technical sections) and mock-ups. |
| Build         | Configuration                                | The built SAP XStep (SapFractal), reusing standard best-practice components where possible.          |
| Qualification | <b>IQ → OQ → PQ</b>                          | Spec <b>Test Scenarios</b> feed OQ; SAP best-practice <b>test scripts</b> are a ready OQ basis (§6). |

**Risk-based testing.** High-risk functions — batch release, quality decisions, e-signatures, calculations that gate disposition — get exhaustive testing; low-risk display/formatting gets lighter scrutiny. Design specs should flag which XStep functions are high-risk so validation effort is proportionate.

**Clean-core reduces validation scope.** Every reused standard XStep / delivered FM is functionality we don't have to validate from scratch. Our "reuse-first, author-new-only-when-nothing-fits" rule is therefore a compliance lever, not just an efficiency one — and any genuinely new Z function module (e.g. in `ZAI_XSTEP_FMG`) is net-new validation scope that the spec must call out.

## 5 SAP PP-PI & the batch-record runtime

- Our EBR lives in **PP-PI (Production Planning for Process Industries)** — SAP's module for recipe/formula manufacturing used by pharma, chemicals and food. The GxP-relevant concepts:
- **Master Batch Record (MBR) / Master Recipe** — the centralised, **versioned** master that drives execution and lays the foundation for standardised, audit-proof, cross-site production. In our world: the SiMPL MBR that overlays recipe values onto XStep `IV_*` parameters, and the **Released** XStep version (the validated one — *not* necessarily the highest-numbered).
  - **Batch Management** — raw materials and finished goods tracked by lot/batch, giving **batch genealogy / traceability** across the lifecycle. Our Part No. / Batch No. columns with expiry validation feed this.
  - **Quality Management (QM)** — inspection planning, sampling, results recording, usage decision, nonconformance/CAPA. Our effluent/results and sampling XSteps are the operator-facing capture points.
  - **Electronic Batch Record & Review by Exception** — the digital record captures deviations and risk classification; reviewers focus only on documented deviations while compliant steps auto-approve — shorter review with full audit assurance. Our in-line range validations and read-only computed fields are what make "by exception" possible.

## 6 SAP Best Practices & the Process Navigator (the reference model)

SAP publishes its "known-good" processes in the **SAP Signavio Process Navigator** (successor to Best Practices Explorer, inside SAP for Me). It's the model we fit-gap against under SAP Activate. Hierarchy:

| Level                                             | Meaning                                                                                                                                                                                  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Solution Scenario</b>                          | A bundle for a solution or end-to-end process (e.g. <i>Plan to Fulfil</i> ).                                                                                                             |
| <b>Solution Process</b> (= scope item, has an ID) | One implementable process. Relevant to us: <b>BJ8 — Make-to-Stock Process Manufacturing Based on Process Order</b> , with a companion <b>QM in Make-to-Stock Process Manufacturing</b> . |
| Solution Process Flow → Activity → Action         | Progressive breakdown to individual user actions.                                                                                                                                        |

Each Solution Process ships **accelerators**: a **BPMN process-flow diagram**, a **test script** (step-by-step execution), and **set-up instructions**. For us, the **test script**  $\approx$  an **OQ test basis** and the process-flow diagram  $\approx$  the process description in a batch record — the same artifacts CSV leans on.

## 7 GxP controls ↔ our XStep building blocks

The core of this document: each regulatory control and the concrete XStep feature that delivers it.

| GxP control                | Requirement                                                               | How our XStep design delivers it                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electronic signatures      | 21 CFR 11 — attributable, secure, unique-ID sign-offs on critical actions | <b>Performed By / Witnessed By</b> signature columns & the Optional-Signature block; signature strategies ( <code>SAPPOCSS</code> ) with validation FMs ( <code>/SMPL/PPPI_FM_VALI_SUPE_SIG</code> , <code>SIG_ADD_DB_CB</code> ). <b>One signature per table line</b> ; second sign-off is a step-level footer. |
| Audit trail                | Time-stamped record of create/modify with user                            | Signature callbacks persist to SAP; FM-stamped date/time ( <code>/SMPL/PPPI_FM_GET_DATE_TIME</code> ) rather than typed values; process messages record consumption with user/date.                                                                                                                              |
| Contemporaneous timestamps | ALCOA — recorded at the time of the act                                   | Date/Time captured by a ► <b>button that stamps a read-only field</b> — never hand-keyed; Date and Time modelled as separate fields.                                                                                                                                                                             |
| Accurate calculations      | No manual arithmetic error; governed formulas                             | Calc XSteps (Three/Four-Variable Calc) and range FMs compute results into <b>read-only computed fields</b> ; standard constants pre-filled read-only, not re-keyed.                                                                                                                                              |
| In-spec enforcement        | Range / tolerance limits enforced, deviations captured                    | Range-validation FMs ( <code>/SMPL/PPPI_FM_MIN_MAX</code> , <code>CALC_TOLERANCE</code> ) on entry columns; MBR-fed <code>IV_*</code> min/max; out-of-spec triggers a deviation/escalation prompt (enables review-by-exception).                                                                                 |
| Controlled vocabularies    | Restricted, consistent data entry                                         | Dropdowns backed by <b>CT04 characteristics</b> ( <code>ZSMPL_CHAR_*</code> ) — a fixed value list, never free text.                                                                                                                                                                                             |
| Material traceability      | Batch genealogy; correct material/batch consumed                          | Part No. → Material Description and Batch No. → Exp. Date <b>input validations</b> ; goods issue via the <b>Z_PICONS process message (movement type 261)</b> , one posting per material line.                                                                                                                    |
| Version / change control   | Annex 11 — controlled configuration & change management                   | XStep <b>Released version</b> is the validated one; changes go through SAP transport / ChaRM; specs are versioned; the design spec is the change-control artifact.                                                                                                                                               |
| Master data integrity      | Mandatory GxP fields present & correct                                    | Equipment/calibration checks via <code>GET_ASSIGNED_EQUI_EBR</code> ; expiry/date validation FMs; MBR-governed setpoints.                                                                                                                                                                                        |
| Reduced transcription      | Fewer human-error opportunities                                           | Carry-forward (read-only) values sourced from upstream XStep outputs; computed fields greyed; one reusable block instantiated many times rather than re-transcribed per occurrence.                                                                                                                              |

## 8 Data integrity (ALCOA+) applied to our design decisions

| Principle | Design decision it drives |
|-----------|---------------------------|
|-----------|---------------------------|

|                              |                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attributable</b>          | Per-line Performed By + footer Witnessed By; signature-validation FMs tie the action to a unique user.                                                                |
| <b>Legible</b>               | Short, standardised column labels ( $\leq \sim 30$ chars $\rightarrow$ <code>PPPI_INPUT_REQUEST</code> ); full description in the instruction text.                   |
| <b>Contemporaneous</b>       | FM-stamped date/time buttons; no post-hoc typing of timestamps.                                                                                                       |
| <b>Original</b>              | Values captured once in the owning step; other steps carry them forward read-only ("don't duplicate data across steps").                                              |
| <b>Accurate</b>              | Computed outputs read-only; standard constants pre-filled read-only; range/tolerance validation on measured inputs.                                                   |
| <b>Complete / Consistent</b> | Coverage crosswalk ensures every MABR section & hidden recordable (totals, cross-records, sub-tables) is captured; sibling "results" tables use a consistent variant. |
| <b>Enduring / Available</b>  | Records persist to SAP (custom <code>ZTC_*</code> tables / standard docs), retrievable for review and audit.                                                          |

## 9 Compliance checklist when authoring an XStep / spec

- Critical actions carry an **e-signature**; one per table line, second to the footer.
- Every **timestamp** is a stamped read-only field (button), Date and Time separate — never typed.
- **Computed / standard-constant** fields are read-only (greyed); only true operator measurements are entry fields.
- Measured inputs with limits carry a **range/tolerance validator**; min/max come from MBR-fed `IV_*`.
- Restricted fields are **dropdowns backed by a CT04 characteristic**, never free text — each is two deliverables (column + characteristic).
- Material lines validate **Part No. / Batch No.** and post goods issue via **Z\_PICONS (261)**, one line per material.
- **All `IV_*` parameters** documented in the spec (the MBR-authorable interface); verify "authored in MBR" claims against the live XStep — some are hardcoded.
- Reuse from the validated library first; **flag genuinely new Z development** as net-new validation scope.
- **Coverage crosswalk** complete — no dropped section, total, cross-record write-back or sub-table.
- Confirm the **Released version** when mirroring an existing XStep.

## 10 Glossary

| Term           | Meaning                                                                                  |
|----------------|------------------------------------------------------------------------------------------|
| eBR / EBR      | Electronic Batch Record — the digital manufacturing record.                              |
| MBR            | Master Batch Record / Master Recipe — the versioned master that drives execution.        |
| PI-PCS / PP-PI | Process Instruction–Process Control System / Production Planning for Process Industries. |
| PI Sheet       | Operator-facing sheet generated from the control recipe; where XSteps render.            |

|                               |                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------|
| XStep                         | Reusable execution-step building block (SAP cmxsvn) that renders into the PI Sheet.        |
| Process message               | Structured message posting data back to SAP (e.g. goods issue Z_PICONS, mvt 261).          |
| CSV                           | Computer System Validation.                                                                |
| IQ / OQ / PQ                  | Installation / Operational / Performance Qualification.                                    |
| GAMP 5                        | ISPE risk-based CSV guidance; SAP ≈ configurable product (Cat 3–4).                        |
| ALCOA+                        | Data-integrity principles (Attributable, Legible, Contemporaneous, Original, Accurate, +). |
| Scope item / Solution Process | One implementable SAP best-practice process (e.g. BJ8).                                    |
| Review by exception           | Reviewing only documented deviations; compliant steps auto-approve.                        |

## 11 Sources & caveats

The controlling SAP page ([me.sap.com/processnavigator](https://me.sap.com/processnavigator)) is behind SAP authentication and could not be retrieved directly; the SAP best-practice structure and GxP mapping below are assembled from public SAP Community / SAP Learning / SAP Help and reputable third-party sources. Verify regulatory specifics against the controlling text before formal use.

1. SAP Signavio Process Navigator — Solution Scenario / Solution Process (SAP Community blog series).
2. SAP Learning: Navigating SAP Best Practice solution processes via SAP Signavio Process Navigator.
3. SAP KBA 3382435 — How to find scope items on SAP Signavio Process Navigator.
4. SAP Community — Quality Management in Process Manufacturing with SAP S/4HANA Cloud; The Ultimate Guide to SAP S/4HANA Batch Management.
5. SAP for Me doc 2964352 — Quality Management in Make-to-Stock Process Manufacturing (scope item BJ8 family).
6. IntuitionLabs — SAP S/4HANA Pharma: GxP Implementation & Validation (21 CFR Part 11, EU Annex 11, GAMP 5, CSV, eBR/MBR).
7. FDA 21 CFR Part 11; EudraLex Vol. 4 GMP Annex 11 & Chapter 4; ISPE GAMP 5 (2nd ed.) — controlling standards.